Project Name: GradPace

Tagline: "Choose your Path – Build your Future"

Dawson Fromm

Old Dominion University

CS410

Professor Rosni

02/06/2026

GradPace – “Choose your Path, Build your Future” – The Career-Planning app

Problem Description

Many college students struggle to understand how their academic progress, extracurricular activities, and professional development fit together into a clear path toward their intended career. This issue is especially common among first-generation students, underclassmen, and students in competitive fields such as computer science, engineering, and business.

Students today rely on multiple disconnected systems:

- Canvas for coursework
- DegreeWorks for academic progress
- Handshake for internships
- LinkedIn for networking
- GitHub for projects
- Email and advising notes for guidance
- …and more

Each platform provides useful information, but they operate in isolation. Students are left to mentally piece everything together to determine whether they are on track for their intended career. As a result, many students:

- Focus primarily on grades without building experience
- Miss internships or research opportunities
- Have incomplete or weak professional profiles
- Choose projects that don’t align with career goals
- Discover gaps in their preparation too late

The lack of real-world experience is a major factor in employment outcomes. Surveys show that **60% of employers avoid hiring recent graduates because they lack practical, real-world experience** (Hult Skills Survey).

At the same time, many students graduate without gaining that experience. Studies indicate that **approximately 50% to 60% of recent college graduates are underemployed or working in roles that do not require their degree**. For the class of 2025, **only about 30% secured entry-level jobs directly related to their field**, representing a five-year low (Cengage Group Employability Report).

These outcomes are not always caused by a lack of ability or effort. Many students simply lack a clear understanding of how their academic choices, projects, internships, and professional presence connect to actual career outcomes.

The problem is not a lack of information. The problem is a **lack of centralization, context,** and **forward-looking guidance**. Students need a system that shows how everything they’re doing fits into a bigger picture. A platform that highlights missing experiences early and gives them clear short-term goals that lead toward long-term employment.

GradPace – “Choose your Path, Build your Future” – The Career-Planning app

Solution

GradPace is a centralized platform that connects academic, professional, and extracurricular data into a clear roadmap for students. Instead of jumping between several disconnected platforms, students can see their entire progress in one place and understand how it aligns with specified career goals.

Key Features

- **Unified Student Dashboard:** Combines academic progress, projects, internships, certifications, LinkedIn presence, and extracurriculars, etc., into one timeline.
- **Goal-Based Path Mapping:** Students select an overarching career goal, such as software engineer, data scientist, or researcher. The system then outlines both long-term and short-term milestones.
  Example:
  - **Long-term goal:** Full-time software engineering job after graduation
  - **Short-term goal (this year):** Secure a summer internship
  - **Supporting steps:**
    - Complete a data structures course
    - Build two portfolio projects
    - Improve LinkedIn and resume

  This structure gives students small, achievable wins, all while keeping them aligned with a larger career objective.

- **Prospective Path Explorer (“What-If” Analysis):** Students can experiment with different career goals and immediately see how their current portfolio aligns with each path.
  Example:
  - **Compare goals:**
    - Software engineering
    - Data science
    - Research or graduate school
  - **System shows:**
    - Which requirements are already met
    - Which skills or experiences are missing
    - The estimated effort needed to transition between paths

  This helps students understand the trade-offs and feasibility before committing to a direction.

- **Gap-Detection System:** Automatically highlights missing or weak areas, such as:
  - No internship experience
  - Weak or incomplete LinkedIn profile
  - Lack of technical projects
  - Missing certifications
  - Limited extracurricular or leadership experience
- **Guided Resource Links:** Instead of directly recommending specific internships or programs, the system will provide targeted guidance.
  Examples:

  - “Internship needed” → link to infographic on tips to find internship opportunities
  - “Research experience recommended” → link to university research page
  - “Certification recommended" → page with recommended certifications and links to official websites
  - “Leadership needed” → link to campus organizations page

This keeps the system scope realistic, while still giving students clear next steps.

- **Progress Visualization:** A visual roadmap showing:
  - Completed milestones
  - Current progress
  - Remaining steps toward the long-term goal
  - Current short-term objective (e.g., “Summer internship”)

Market Potential

Who are the users?

Primary users include:

- First-generation college students
- Underclassmen unsure of career direction
- Students in competitive or rapidly evolving fields
- Students balancing coursework, projects, and internships

There are over **19 million college students in the United States** (Education Data Initiative), many of whom face similar challenges in navigating academic and career planning. There is a large and growing population of students who want better guidance and clearer direction during college. *See Figure 1. Student Perception of Career Readiness*

Who are the customers?

Potential customers could include universities, advising departments, career services offices, and educational technology companies. Scholarship programs and internship providers could also benefit from the platform. Institutions could license GradPace to improve career placement rates, support advising, help students make better academic and professional decisions, and demonstrate stronger post-graduation outcomes.

Who are the stakeholders?

Key stakeholders include students seeking career clarity, universities trying to improve employment outcomes, employers seeking better-prepared graduates, and advisors or mentors who support students through their academic journey.

Visual Aid:

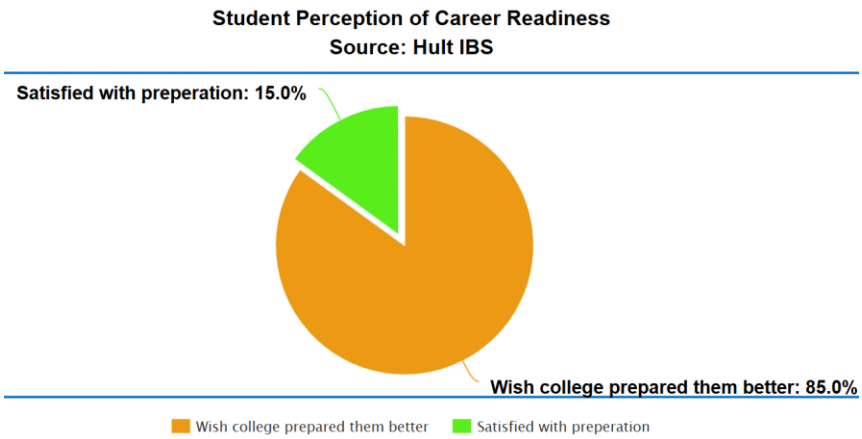

Figure 1. *85% of recent graduates say they wish their undergraduate education had better prepared them for the workplace, indicating strong demand for tools that help students align their college experience with career outcomes (Hult Skills Survey)*

Competitive Landscape and Challenges

Competitors

Existing platforms address only parts of the problem. DegreeWorks focuses on academic progress, Handshake provides internship and job listings, and LinkedIn supports professional networking. Personal planning tools such as Notion or traditional planners help students stay organized. All these platforms work great in isolation but struggle to be cohesive. **GradPace** would offer a unified academic/professional experience, implementing a career roadmap, goal-based planning, prospective “what-if” analysis, and automated gap detection across multiple platforms.

What makes GradPace better?

GradPace focuses on **career alignment**, not just information. Instead of showing isolated data, it connects all aspects of a student’s development and shows how each decision contributes to specific career paths. It allows students to compare different possible futures, identify missing experience early, and get suggested next steps. In this way, it functions more like a career navigation system than a simple planner or job board.

Challenges to overcome

A primary challenge would be **data integration**, since pulling information from multiple platforms may require APIs and/or manual input. Another challenge is building a user base, as students may be hesitant to use another tool unless it provides obvious value.

Additionally, institutional partnerships may take time to establish, as universities usually require approvals and demo programs before adopting new systems. Nevertheless, these challenges can be managed through gradual implementation, where we could start with a simple standalone version that proves its value before deeper integrations are attempted.

Conclusion

GradPace turns uncertainty into direction. Instead of leaving students to guess whether they’re on track, it provides a clear roadmap that connects their coursework, experiences, and professional development to real career outcomes. By breaking long-term ambition into short-term, achievable goals, the platform gives students consistent momentum to build on throughout their college careers.

References

Brown, E. (2025, January 21). *New survey reveals traditional undergraduate education is not preparing students for the workforce* . Hult International Business School. https://www.hult.edu/blog/wi_skills_survey/

Employability Report - Annual graduate employability reports. Cengage Group. (n.d.). https://www.cengagegroup.com/edtech-research/employability-report/

Hanson, M., & Checked, F. (2025, March 17). *College enrollment statistics [2026]: Total + by demographic*. Education Data Initiative. https://educationdata.org/college-enrollment-statistics